Math 74: Algebraic Topology

Prishita Dharampal

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Problem 1. Show that if $h, h' : X \to Y$ are homotopic and $k, k' : Y \to Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ is are homotopic.

Solution.

Assume $h \simeq h'$ and $k \simeq k'$. Then there exist F_h and F_k such that

$$F_h(x, 0) = h(x),\ F_h(x, 1) = h'(x),\ F_k(x, 0) = k(x),\ F_k(x, 1) = k'(x)$$

To show that $k \circ h \simeq k' \circ h$, we define $F : X \times I \to Z$ as,

$$F(x, t) = F_k(F_h(x, t), t)$$

Note that F is the composition of continuous maps and hence is continuous. We verify initial and final points.

$$F(x, 0) = F_k(F_h(x, 0), 0) = F_k(h(x), 0) = k(h(x)) = k \circ h(x)$$

$$F(x, 1) = F_k(F_h(x, 1), 1) = F_k(h'(x), 1) = k'(h'(x)) = k' \circ h'(x)$$

Therefore there exists a continuous deformation of $k \circ h$ to $k' \circ h'$, and $k \circ h \simeq k' \circ h$.

Problem 2. A space X is said to be contractible if the identity map $i_x : X \to X$ is homotopic to a constant map.

1. Show that I and R are contractible.
2. Show that a contractible space is path connected.
3. Show that if Y is contractible, then for any X the set $[X, Y]$ has a single element.
4. Show that if X is contractible and Y is path connected, then $[X, Y]$ has a single element.

Solution.

1.

Problem 3. A subset A of $\mathbb{R}^n$ is said to be star convex if for some point a_0 of A, all the line segments joining a_0 to other points of A lie in A.

1. Find a star convex set that is not convex.
2. Show that if A is star convex, A is simply connected.
3. Show that if A is star convex, any two paths in A having the same initial and final points are path homotopic.

Problem 4. Let x_0 and x_1 be two given points of the path-connected space X. Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair α and β of paths from x_0 to x_1, we have $\hat{\alpha} = \hat{\beta}$.

Problem 5. Let A be a subset of $\mathbb{R}^n$ let $h : (A, a_0) \to (Y, y_0)$. Show that if h is extendable to a continuous map of $\mathbb{R}^n$ into Y, then h_* is the zero homomorphism (the trivial homomorphism that maps everything to the identity element).

Problem 6. Let G be a topological group with operation $\cdot$ and the identity element x_0. Let $\Omega(G, x_0)$ denote the set of all loops in G based at x_0. If $f, g \in \Omega(G, x_0)$, let us define a loop $f \otimes g$ by the rule

$$(f \otimes g)(s) = f(s) \cdot g(s)$$

1. Show that this operation makes the set $\Omega(G, x_0)$ into a group.

2. Show that this operation induces a group operation $\otimes$ on $\pi_1(G, x_0)$.

3. Show that the two group operations $\star$ and $\otimes$ on $\pi_1(G, x_0)$ are the same. [*Hint:* Compute $(f \star e_{x_0}) \otimes (e_{x_0} \star g)$].

4. Show that $\pi_1(G, x_0)$ is abelian.

Problem 7. Show that the map $f : S^1 \to S^1$, defined as $f(z) = z^n$ for $z \in S^1 \subset \mathbb{C}$, is a covering map. Here we think of S^1 as $\{z \in \mathbb{C},\ \text{such that } |x| = 1\}$.